Announced Rescue Payments with Strategic Drivers

Commitment, Universal Rejection, and Latent Local Supply

Abstract

We study a platform that commits ex ante to a payment for a request and a higher payment that becomes available only after universal rejection. The realized local driver pool is Poisson and unobserved; private-cost incumbents simultaneously choose whether to accept now or preserve a discounted option to wait. The rider privately decides whether to post and, after failure, whether to abandon, repeat, or activate rescue. We first characterize every anonymous symmetric pure-cutoff weak perfect Bayesian equilibrium under an arbitrary menu, including beliefs and boundary actions. Whenever a positive rider mass uses marginal rescue, a small announced increase strictly raises completion for every fresh-entry rate. In the no-entry baseline, we prove the stronger result that every menu has a unique cutoff. This yields an exact solution of the platform's full policy problem: conditional on the first payment, the optimal rescue payment is either a reject-all boundary, a unique tangent rescue, or an irrelevant repeat-only payment. A Lambert-W envelope reduces the global two-price problem to a continuous scalar maximum, so an optimal menu exists. The dynamic policy strictly dominates the optimized flat payment whenever rider patience exceeds an endogenous threshold; in particular, it does so for every finite market thickness when $\beta \geq 1/2$. The gain is continuous in thickness and has an attained interior maximum. Driver impatience changes the thin-market order: for $\beta > 1/2$ the gain is first order in m when drivers strictly discount, but only second order at the no-discount boundary. The thick-market rate remains $(\log m)/m$. We do not prove strict single-peakedness; when $\beta < 1/2$, an initial zero plateau rules it out. A public-N benchmark separates two local mechanisms: intertemporal compensation already operates with one incumbent, while assignment competition amplifies it with several.

1 Introduction

Platforms often increase the payment attached to a request that remains unfilled. If the increase is announced in advance, it creates a commitment problem. A higher rescue payment makes a failed request easier to fill, but it also makes rejection today more attractive. Completion design must therefore account for the strategic delay induced by the rescue policy itself.

We study a single request and two periods. Before any private action, the platform announces a menu (p_1, p_2) with $p_2 \geq p_1$. A rider with private value decides whether to post. Private-cost incumbent drivers then simultaneously accept or wait. Waiting discounts an incumbent's terminal net payoff by $\delta \in (0, 1]$. If at least one accepts, one is selected uniformly. If all reject, the rider observes only this public failure and may abandon, repeat p_1, or activate p_2. Incumbents can survive and fresh drivers can enter before the terminal acceptance decision.

The information structure is central. Market thickness m is public, but the realized incumbent count is not observed by the platform, rider, or any individual driver. Conditional on a focal driver being present, her rival count remains Poisson by Palm conditioning. Universal rejection consequently does two jobs: it activates rescue and reveals that no incumbent below the equilibrium cutoff accepted. A driver's assignment probability after accepting now is

$$\phi(ma) = \frac{1 - e^{-ma}}{ma},$$

whereas reaching the continuation game after waiting requires every rival to reject, an event with probability e^{-ma}. Assignment competition is one operational wedge behind the local effect. A second wedge is temporal: announced escalation can compensate a waiting incumbent for discounting. Neither wedge by itself is our mathematical contribution.

Our analysis follows the policy–response–optimization order used by Aviv and Pazgal (2008): fix an announced policy, solve the strategic threshold response, and only then optimize. The economic direction is reversed. Forward-looking consumers in announced-markdown models wait for a lower price and risk stockout; our suppliers wait for a higher payment and risk losing the request to a rival.

The paper makes four contributions. First, it gives a complete cutoff-WPBE characterization under arbitrary survival and entry, with the Palm posterior, failure posterior, rider continuation choices, boundary cutoffs, and tie-breaking stated explicitly. The assessment is conservative only within the anonymous symmetric pure-cutoff class; it is not a worst case over every mixed or asymmetric PBE. We then prove a uniform local result across fresh entry rates: if marginal rescue is active for positive rider mass, the unique nearby cutoff moves according to an explicit coefficient and completion rises strictly.

Second, for the uniform no-entry baseline we solve the global menu problem. Every strict active menu has one cutoff, not merely a locally selected branch. For a fixed p_1, rescue-price design becomes a strictly concave envelope. We identify the exact reject-all, tangent-rescue, and repeat-only regimes and then invert the fixed-price solution with the feasible root of one quadratic. The outer problem is a continuous one-dimensional maximum, which proves policy attainment and gives an exact optimizer correspondence.

Third, we characterize rider patience and market thickness. Strict dynamic gain occurs exactly for $\beta > \beta_c(m, \alpha, \delta)$, where the threshold has an independently checkable scalar test and explicit bounds. Since the unique flat optimizer is below $1/2$, the sufficient condition extends to $\beta = 1/2$. In thickness, the optimized gain is continuous, vanishes at both endpoints, and attains a positive interior maximum. We prove optimized endpoint rates rather than transferring a local derivative through an optimization. We deliberately do not call this single-peaked: the required global derivative theorem is unavailable, and for $\beta < 1/2$ strict single-peakedness is false because the value is initially zero.

Fourth, a public deterministic-supply benchmark separates latent information, driver impatience, and assignment competition. With $\delta < 1$, a marginal rescue strictly raises completion even for one known incumbent. At $\delta = 1$, that temporal term disappears: the one-incumbent derivative is zero and the derivative is positive exactly when $n \geq 2$. Latent Poisson supply shapes inference and global design, but is not required for either local mechanism.

2 Related literature and contribution boundary

We do not claim to introduce multistage bonuses or failure-contingent payment increases. The closest operational work is Wu et al. (2022), who optimize multistage order-specific bonuses in meal delivery and document substantial reductions in cancellation. Their stage acceptance model depends on the current bonus; it does not model a driver who strategically responds to an ex ante announced future bonus. Two-round broadcast and notification mechanisms also study a second attempt after no response, but vary reach or assignment rather than solve an announced same-request wage game (Sun et al., 2020; Qin et al., 2025).

The closest behavioral work studies strategic supplier rejection. Sigg et al. (2025) model workers who reject a delivery request so that it is reoffered at a higher payment, emphasizing labor oversupply and collective action. Chen (2012) derives private-cost sellers' willingness to wait for higher buyer offers when the seller count is fixed and known and the buyer does not precommit. Related work examines failure-contingent rewards, rejection in anticipation of future surge, and strategic withholding of available supply (Acemoglu et al., 2014; Garg and Nazerzadeh, 2022; Bai et al., 2023). Our narrower object is the intersection of ex ante commitment, simultaneous same-request competition, an unobserved realized rival pool, and universal rejection as an endogenous public signal.

Finally, two-sided dynamic pricing already incorporates forward-looking buyers and sellers. Chen and Hu (2020) study Poisson arrivals and strategic waiting in a dynamic matching market. Hu et al. (2022) obtain both high–low and low–high two-period surge paths under rider and driver temporal responses. Announced and contingent pricing with strategic consumers provides the methodological mirror (Aviv and Pazgal, 2008; Correa et al., 2016). Relative to these papers, our policy is attached to one request, available only after universal rejection, and optimized over the equilibrium delay it induces among the same incumbents.

3 Model

3.1 Primitives and policy

There is one potential request and two periods. Market thickness $m > 0$ is public. Before rider or driver private information is used, the platform commits to

$$\mathcal{P}^D = \{(p_1, p_2) : 0 \leq p_1 \leq p_2 \leq 1\}.$$

The second payment is available only following first-period failure and cannot be revised ex post. A flat policy is (p, p). Payment is a transfer from the rider to the selected driver. The platform maximizes the unconditional probability that the potential request is completed; it does not maximize profit or welfare.

The rider has $v \sim U[0,1]$. A period-1 completion yields $v - p_1$ and a period-2 completion at payment p_j yields $\beta v - p_j$, where $\beta \in (0, 1)$. Incumbents have a separate discount factor $\delta \in (0, 1]$: if an incumbent waits and is selected in period 2, her period-1-equivalent net payoff is $\delta(p_j - c)$. A fresh entrant arrives only in period 2 and evaluates the contemporaneous payoff $p_j - c$. Thus β measures rider patience and δ measures incumbent-driver patience; the common-discount specification is the restriction $\delta = \beta$. From the request perspective the incumbent count is

$$N^I \sim \text{Pois}(m).$$

The realization is latent to every player. Incumbent costs are i.i.d. $U[0, 1]$ and remain fixed across periods. An accepting driver incurs her cost only if selected; an unselected acceptor earns and incurs zero.

After universal rejection, each incumbent survives independently with probability $\alpha \in [0, 1]$. Independently, a fresh pool

$$N^E \sim \text{Pois}(\gamma m), \qquad \gamma \geq 0,$$

arrives with i.i.d. uniform costs. The rider observes neither counts nor survival. At the terminal node, an available driver accepts active payment q exactly when $c \leq q$.

3.2 Timing and information

Figure 1 gives the reduced extensive form. The driver nodes represent simultaneous Bayesian moves, not observation of a realized pool.

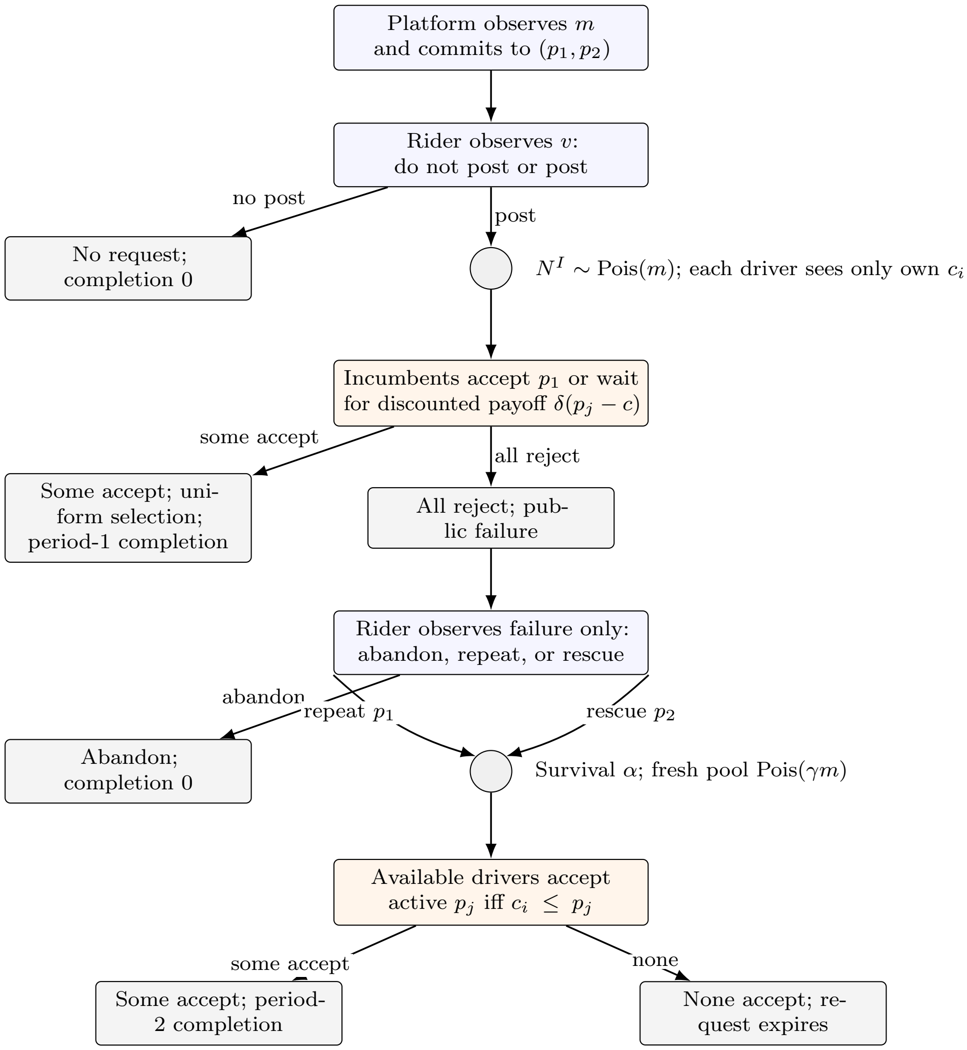

Figure 1: Reduced extensive form. The realized incumbent count, rival costs, survival, and fresh entry remain latent at the indicated decisions.

3.3 Equilibrium class, beliefs, and tie-breaking

We use weak perfect Bayesian equilibrium (WPBE), restricted to anonymous, symmetric, pure first-period driver cutoffs. Under cutoff a, incumbents below a accept and those above a wait. The cutoff type accepts at an interior or upper cutoff and rejects at the lower cutoff; this affects only a null cost type.

The rider posts at a zero payoff when $v = p_1$, and does not post below it. After failure, the following priority rule makes continuation single-valued. If the maximal continuation payoff is zero, the rider repeats iff $\beta v - p_1 \geq 0$ and otherwise abandons. If repeat and rescue tie at a strictly positive maximum, the rider chooses rescue. Terminal drivers accept at a zero surplus. These rules matter when repeat has zero coverage: a positive interval of rider types can then be indifferent between repeat and abandon.

For $p_1 < 1$, posting implies the belief $v \mid \{\text{post}\} \sim U[p_1, 1]$. Since supply is independent of value, failure leaves this value posterior unchanged. Poisson splitting determines the rider's supply posterior after failure. For a focal incumbent, Palm conditioning leaves the number of rivals distributed as $\text{Pois}(m)$. At $(p_1, p_2) = (1, 1)$, posting is a null event; we complete the off-path driver subgame with cutoff one and define completion as zero.

4 Equilibrium under an arbitrary announced menu

Fix a candidate cutoff $a \in [0, p_1]$ and define

$$\phi(x) = \begin{cases} (1 - e^{-x})/x, & x > 0, \\ 1, & x = 0, \end{cases} \qquad \lambda_j(a) = m\{\alpha(p_j - a) + \gamma p_j\}, \qquad C_j(a) = 1 - e^{-\lambda_j(a)}. \tag{1}$$

Lemma 1 (Poisson competition and failure posterior)**.** *Under cutoff a, first-period completion conditional on posting is $1 - e^{-ma}$. A focal acceptor's expected assignment share is $\phi(ma)$, while a focal waiting driver reaches failure against all rivals with probability e^{-ma}. Conditional on universal rejection, incumbents above a remain a Poisson marked population. After survival and fresh entry, terminal coverage at p_j is $C_j(a)$, and an eligible focal driver's terminal assignment share is $\phi(\lambda_j(a))$.*

The assignment identity follows from

$$\mathbb{E}\left[\frac{1}{1+Z}\right] = \int_0^1 \mathbb{E}[t^Z]dt = \frac{1 - e^{-x}}{x}, \qquad Z \sim \text{Pois}(x).$$

Poisson marking gives independent counts below and above a. Conditional on no point below a, the above-a process is unchanged. This is not the law of $N^I - 1$ conditional on $N^I \geq 1$.

After failure, rider payoffs from abandon, repeat, and rescue are

$$0, \qquad C_1(a)(\beta v - p_1), \qquad C_2(a)(\beta v - p_2).$$

If $C_2 > C_1$, define

$$v^M(a) = \frac{C_2(a)p_2 - C_1(a)p_1}{\beta[C_2(a) - C_1(a)]};$$

if $C_2 = C_1$, put $v^M = +\infty$. Conditional on posting, the repeat and rescue masses are

$$\eta^R(a) = \frac{[\min\{v^M(a), 1\} - p_1/\beta]^+}{1 - p_1}, \qquad \eta(a) = \frac{[1 - v^M(a)]^+}{1 - p_1}, \tag{2}$$

and

$$\eta^R(a) + \eta(a) = \rho(p_1) := \frac{[1 - p_1/\beta]^+}{1 - p_1}. \tag{3}$$

The initial posting payoff is

$$(1 - e^{-ma})(v - p_1) + e^{-ma} \max\{0, C_1(\beta v - p_1), C_2(\beta v - p_2)\},$$

so the rider posts exactly when $v \geq p_1$.

A focal incumbent of cost c obtains

$$A(c; a) = \phi(ma)(p_1 - c) \tag{4}$$

from accepting immediately and

$$W(c;a) = \delta\alpha e^{-ma}\left\{\eta^R(a)\phi(\lambda_1(a))[p_1 - c]^+ + \eta(a)\phi(\lambda_2(a))[p_2 - c]^+\right\} \tag{5}$$

from waiting. These are conditional on observing a posted request; the posting mass must not multiply a driver's sequential payoff.

For $c \leq p_1$, the margin $A - W$ is affine and strictly decreasing in c:

$$A(c;a) - W(c;a) = B(a)(p_1 - c) - D(a), \tag{6}$$

where

$$\begin{aligned} B(a) &= \phi(ma) - \delta\alpha e^{-ma}\{\eta^R(a)\phi(\lambda_1) + \eta(a)\phi(\lambda_2)\}, \\ D(a) &= \delta\alpha e^{-ma}\eta(a)\phi(\lambda_2)(p_2 - p_1). \end{aligned}$$

Indeed, $B(a) > 0$ for $p_1 > 0$ because

$$B(a) \geq \phi(ma) - \delta e^{-ma} \geq 0,$$

with strictness from $a > 0$ or, at $a = 0$, from $\rho(p_1) < 1$. Hence a best response is genuinely a cutoff.

Define the cutoff-type margin

$$f_m(a;p_1,p_2) = \phi(ma)(p_1 - a) - \delta\alpha e^{-ma}\left\{\eta^R(a)\phi(\lambda_1)(p_1 - a) + \eta(a)\phi(\lambda_2)(p_2 - a)\right\}. \tag{7}$$

Proposition 1 (Symmetric cutoff WPBE). *For $0 < p_1 < 1$, the equilibrium cutoff set is*

$$\begin{aligned} \mathcal{E}_m^{\alpha,\delta,\gamma}(p_1,p_2) = \{0 &: f_m(0;p_1,p_2) \leq 0\} \\ &\cup \{a \in (0,p_1) : f_m(a;p_1,p_2) = 0\} \\ &\cup \{p_1 : f_m(p_1;p_1,p_2) \geq 0\}. \end{aligned} \tag{8}$$

It is nonempty and compact. At $p_1 = 0$, the only aggregate cutoff is zero with the lower-boundary reject action; at $(1,1)$, set $\mathcal{E}_m^{\alpha,\delta,\gamma}(1,1) = \{1\}$. Thus every feasible menu admits an anonymous symmetric pure-cutoff WPBE.

At the upper boundary,

$$f_m(p_1;p_1,p_2) = -\delta\alpha e^{-mp_1}\eta(p_1)\phi(\lambda_2(p_1))(p_2 - p_1) \leq 0,$$

so $a = p_1$ is eligible precisely when the right side is zero. Continuity of f_m, the lower-boundary test, and the intermediate value theorem give existence; endpoint limits remain eligible, proving compactness.

For $p_1 < 1$, completion on cutoff branch a is

$$M_m^{\alpha,\gamma}(p_1,p_2;a) = (1 - p_1)\left[1 - e^{-ma} + e^{-ma}\{\eta^R(a)C_1(a) + \eta(a)C_2(a)\}\right]. \tag{9}$$

We define the conservative value within the stated class by

$$\underline{M}_m^{\alpha,\delta,\gamma}(p_1,p_2) = \min_{a \in \mathcal{E}_m^{\alpha,\delta,\gamma}(p_1,p_2)} M_m^{\alpha,\gamma}(p_1,p_2;a), \tag{10}$$

which is well-defined by Proposition 1. The announced value is

$$D^*_{\alpha,\delta,\gamma}(m) = \sup_{(p_1,p_2) \in \mathcal{P}^D} \underline{M}_m^{\alpha,\delta,\gamma}(p_1,p_2). \tag{11}$$

We retain a supremum here; attainment will be proved for the main no-entry baseline rather than assumed in the general model.

5 Flat-payment benchmark

Proposition 2 (Flat policy)**.** *For every $m > 0$ and $p > 0$, (p, p) has the unique numeric cutoff $a = p$. Its completion is*

$$Q^F_\gamma(m, p) = (1 - p)(1 - e^{-mp}) + e^{-mp}[1 - p/\beta]^+(1 - e^{-\gamma mp}). \tag{12}$$

Consequently

$$F^*_\gamma(m) = \max_{p\in[0,1]} Q^F_\gamma(m, p) \tag{13}$$

is attained.

For $a < p$, the flat cutoff margin factors as

$$(p - a) \left[\phi(ma) - \delta\alpha e^{-ma}\rho(p)\phi\big(m\{\alpha(p - a) + \gamma p\}\big)\right] > 0,$$

so no lower cutoff is sustainable. At $a = p$, surviving rejectors have cost above p and do not accept the same payment later; only fresh drivers can complete after failure, which gives (12). Continuity on $[0, 1]$ gives the maximum in (13).

6 Local escalation with fresh entry

The general equilibrium characterization permits a sharp local comparison even when fresh drivers enter. Fix $0 < p < \beta$ and abbreviate

$$x = mp, \quad R = e^{-x}, \quad E = e^{-\gamma x}, \quad \sigma = \phi(x), \quad \ell = \phi(\gamma x), \quad \rho = \frac{1 - p/\beta}{1 - p}. \tag{L1}$$

For $\alpha + \gamma > 0$, define the limiting repeat–rescue switch and rescue mass by

$$\bar{v} = \frac{1}{\beta}\left[p + \frac{1 - E}{mE(\alpha + \gamma)}\right], \qquad \eta^0 = \frac{1 - \bar{v}}{1 - p}. \tag{L2}$$

At $\gamma = 0$, the fraction in (L2) is zero.

Theorem 1 (Fresh-entry local rescue)**.** *Suppose $m > 0$, $0 < p < \beta < 1$, $0 < \delta \le 1$, $0 \le \alpha \le 1$, $\gamma \ge 0$, $\alpha + \gamma > 0$, and $\bar{v} < 1$. For all sufficiently small $\varepsilon > 0$, the menu $(p, p + \varepsilon)$ has a unique anonymous symmetric cutoff WPBE, with*

$$a_\varepsilon = p - \kappa\varepsilon + o(\varepsilon), \qquad \kappa = \frac{\delta R\alpha\ell\eta^0}{\sigma - \delta R\alpha\rho\ell}. \tag{L3}$$

Every candidate cutoff is uniformly localized by

$$0 \le p - a_\varepsilon \le \frac{\delta\alpha\rho}{1 - \delta\alpha\rho}\varepsilon. \tag{L4}$$

The right derivative of completion is

$$\left.\frac{dM^{\alpha,\gamma}_m(p, p + \varepsilon; a_\varepsilon)}{d\varepsilon}\right|_{0+} = m(1 - p)R\eta^0 \frac{T_\delta}{\sigma - \delta R\alpha\rho\ell} > 0, \tag{L5}$$

where

$$T_\delta = E(\alpha + \gamma)\sigma - \delta R\alpha\ell\{1 - \rho + \rho E(1 + \gamma)\} > 0. \tag{L6}$$

The activity condition $\bar{v} < 1$ says that a positive mass of riders uses marginal rescue after failure. It is automatic at $\gamma = 0$ but substantive with entry. The knife edge $\bar{v} = 1$ needs a second-order rider-mass expansion and is not included. For $\gamma > 1$, activity cannot simply be deleted from the sign claim: for example

$$x = 1, \quad p = \tfrac{1}{2}, \quad m = 2, \quad \beta = \tfrac{51}{100}, \quad \alpha = 1, \quad \gamma = 10, \quad \delta = \tfrac{1}{2}$$

has $\bar{v} > 1$ and makes T_δ in (L6) negative. Thus the theorem's restriction is economically and mathematically real.

7 Global menu design without fresh entry

We now solve the primary baseline

$$\gamma = 0, \qquad m > 0, \qquad \alpha \in (0, 1], \qquad \beta \in (0, 1), \qquad \delta \in (0, 1].$$

Write $(p, q) = (p_1, p_2)$ and $k = \alpha m$. Menus with $p \geq \beta$ or $q \geq \beta$ cannot induce positive-mass rescue and are outcome-equivalent to the flat first payment. The nontrivial region is therefore $0 \leq p < q < \beta$.

For $a \in [0, p]$, put

$$C(x) = 1 - e^{-kx}, \qquad h(a) = \begin{cases} (e^{ma} - 1)/a, & a > 0, \\ m, & a = 0, \end{cases} \tag{14}$$

and

$$A_z(a) = (\beta - z)C(z - a), \quad z \in \{p, q\}, \qquad B_p = \beta(1 - p). \tag{15}$$

The repeat–rescue switch lies below one exactly when $A_q(a) > A_p(a)$. Substitution of the switch type into continuation coverage gives the exact cancellation

$$K_{p,q}(a) = \frac{\max\{A_p(a), A_q(a)\}}{B_p}. \tag{16}$$

Multiplying the cutoff driver's payoff margin by the positive factor me^{ma} gives the sign-equivalent scalar

$$\mathcal{D}^{\delta}_{p,q}(a) = (p - a)h(a) - \delta K_{p,q}(a). \tag{17}$$

The repeat branch cannot contain a cutoff below p:

$$(p - a)h(a) > \frac{A_p(a)}{B_p} \geq \delta \frac{A_p(a)}{B_p}, \qquad 0 \leq a < p. \tag{18}$$

For $a > 0$, this is $\phi(ma) > e^{-ma}$ combined with $\alpha\rho\phi(k(p - a)) \leq 1$; at $a = 0$, strictness follows from $\rho(p) < 1$. Hence every zero of (17) under a strict active menu lies on the rescue branch, not at the rider kink.

Theorem 2 (Global cutoff uniqueness)**.** *Fix $0 \leq p < q < \beta$. If $\mathcal{D}^{\delta}_{p,q}(0) \leq 0$, the unique numeric cutoff is the lower-boundary cutoff $a = 0$. If $\mathcal{D}^{\delta}_{p,q}(0) > 0$, there is a unique cutoff $a \in (0, p)$. Equality at zero never coexists with a positive root. Together with the flat, repeat-only, $p = 0$, and $\alpha = 0$ boundary cases, every no-entry menu has a unique numeric symmetric cutoff under the stated cutoff-type convention.*

The one-crossing argument is global. At an active-rescue zero write $u = p - a$, $z = q - a$, $b = \beta - q$, and $B = \beta(1 - p)$. The root equation is

$$uh(a) = \delta \frac{b}{B} C(z).$$

Since $B - b = q - \beta p > 0$ and $h(a) \geq m$, $u < C(z)/m$. For $R(a) = uh(a)/C(z)$,

$$\frac{R'(a)}{R(a)} = -\frac{1}{u} + \frac{h'(a)}{h(a)} + \frac{k}{e^{kz} - 1} < 0, \tag{19}$$

because

$$\frac{1}{u} - \frac{k}{e^{kz} - 1} > m \frac{1 - \alpha e^{-kz}}{1 - e^{-kz}} \geq m > \frac{h'(a)}{h(a)}.$$

Thus every zero crosses downward. Since $\mathcal{D}^{\delta}_{p,q}(p) < 0$, two zeros would require an intervening upward crossing. At $p = 0$, the numeric cutoff is zero and the cost-zero type uses the lower-boundary reject action.

7.1 Completion identity and comparison with the same flat payment

At any positive cutoff, indifference gives

$$e^{-ma} K_{p,q}(a) = \frac{m\phi(ma)(p - a)}{\delta}.$$

Using $1 - e^{-ma} = ma\phi(ma)$ in (9) yields

$$\boxed{M^{\alpha,\delta,0}_{m}(p, q; a) = (1 - p)m\phi(ma) \left[a + \frac{p - a}{\delta}\right], \qquad a > 0.} \tag{20}$$

At a reject-all equilibrium,

$$M^{\alpha,\delta,0}_{m}(p, q; 0) = \left(1 - \frac{q}{\beta}\right)(1 - e^{-kq}) \geq \frac{(1 - p)mp}{\delta}, \tag{21}$$

with equality exactly at the lower-boundary indifference condition. Repeat may be used by positive mass in (21); its contribution cancels algebraically.

Both $\phi(ma)$ and $a + (p - a)/\delta$ weakly decrease in $a > 0$, and the first decreases strictly, so (20) decreases with the cutoff. Theorem 2 then gives a particularly strong comparison. For every $0 < p < q < \beta$, the strict-menu cutoff is below p, and therefore

$$\underline{M}^{\alpha,\delta,0}_{m}(p, q) > F_m(p) = (1 - p)(1 - e^{-mp}). \tag{22}$$

This is a global same-p result, not merely a derivative along one selected branch.

For completeness, a one-sided local expansion at (p, p) is

$$a_\varepsilon = p - \kappa\varepsilon + o(\varepsilon), \qquad \kappa = \frac{\delta e^{-mp}\alpha\rho}{\phi(mp) - \delta e^{-mp}\alpha\rho}, \tag{23}$$

and

$$\left.\frac{dM(p, p + \varepsilon)}{d\varepsilon}\right|_{0+} = \frac{m(1 - p)e^{-mp}\alpha\rho[\phi(mp) - \delta e^{-mp}]}{\phi(mp) - \delta e^{-mp}\alpha\rho} > 0. \tag{24}$$

The wedge decomposes as $\phi(mp) - \delta e^{-mp} = [\phi(mp) - e^{-mp}] + (1 - \delta)e^{-mp}$: assignment competition is the first term and incumbent impatience the second.

7.2 The fixed-first-payment problem

Following the design sequence, fix p before optimizing q. For a target cutoff $a < \beta$, maximize

$$A(a, q) = (\beta - q)[1 - e^{-k(q-a)}], \qquad q \in [a, \beta]. \tag{25}$$

It is strictly concave. Its unique maximizer $Q(a)$ satisfies

$$e^{k(Q(a)-a)} - 1 = k(\beta - Q(a)), \tag{26}$$

or

$$Q(a) = \beta - \frac{W_0(e^{1+k(\beta-a)}) - 1}{k}. \tag{27}$$

Moreover

$$a < Q(a) < \frac{a + \beta}{2}.$$

Let

$$S(a) = A(a, Q(a)), \qquad S(a) = \frac{k[\beta - Q(a)]^2}{1 + k[\beta - Q(a)]}. \tag{28}$$

Define $Q_0 = Q(0)$, $S_0 = S(0)$, and

$$p_z = \frac{1 - \sqrt{1 - 4\delta S_0/(\beta m)}}{2}. \tag{29}$$

The strict flat inequality at $p = Q_0, a = 0$ gives $0 < p_z < Q_0 < 1/2$.

Theorem 3 (Optimal rescue conditional on the first payment)**.** *Let*

$$H_m(p) = \max_{q \in [p,1]} \underline{M}_m^{\alpha,\delta,0}(p, q).$$

Then:

(i) If $0 \le p \le p_z$, the unique optimal rescue is $q = Q_0$, the cutoff is the lower-boundary reject-all cutoff $a = 0$, and

$$H_m(p) = S_0/\beta.$$

(ii) If $p_z < p < \beta$, there is a unique $a^*(p) \in (0, p)$ satisfying

$$(p - a)h(a) = \frac{\delta S(a)}{\beta(1 - p)}. \tag{30}$$

The unique optimal rescue is $q = Q(a^*(p))$, and

$$H_m(p) = (1 - p)m\phi(ma^*(p)) \left[a^*(p) + \frac{p - a^*(p)}{\delta}\right].$$

(iii) If $\beta \le p \le 1$, every $q \in [p, 1]$ is outcome-equivalent and

$$H_m(p) = (1 - p)(1 - e^{-mp}).$$

Thus the conditional supremum is a maximum in every regime.

To see implementability, a positive target cutoff solves

$$A(a,q) = \frac{\beta(1-p)(p-a)h(a)}{\delta}. \tag{31}$$

The right side strictly exceeds $A(a,p)$ by (18), so (31) cannot generate a repeat-only or kink pseudo-solution. If the envelope is below, equal to, or above the right side, there are respectively zero, one tangent, or two strict implementing prices. Theorem 2 makes each implemented cutoff the unique equilibrium. Completion falls with the cutoff, so the smallest implementable cutoff, attained at tangency, solves the fixed-p problem.

7.3 One-dimensional global design and attainment

Define

$$R_\delta(a) = \frac{\delta S(a)}{\beta h(a)}.$$

The tangent equation is

$$(1-p)(p-a) = R_\delta(a). \tag{32}$$

Because $Q(a) < (1+a)/2$ and the left side is increasing in p below that vertex, the unique feasible root is the lower quadratic branch

$$\boxed{P_\delta(a) = \frac{1+a-\sqrt{(1-a)^2-4R_\delta(a)}}{2}.} \tag{33}$$

The upper branch is not a fixed-p rescue optimum. The map P_δ is continuous and strictly increasing from $[0,\beta]$ onto $[p_z,\beta]$. Endpoint injectivity follows from the same envelope one-crossing argument: $P_\delta(a) = P_\delta(0) = p_z$ for $a > 0$ would give the fixed-p_z envelope two zeros.

Extend $P_\delta(\beta) = Q(\beta) = \beta$ and set

$$J_\delta(a) = \begin{cases} (1-P_\delta(a))m\phi(ma)\left[a + \dfrac{P_\delta(a)-a}{\delta}\right], & a > 0, \\ mp_z(1-p_z)/\delta = S_0/\beta, & a = 0. \end{cases} \tag{34}$$

Then $J_\delta(\beta) = (1-\beta)(1-e^{-m\beta})$, and the complete outer problem is

$$\boxed{D^*_{\alpha,\delta,0}(m) = \max\left\{\max_{a\in[0,\beta]} J_\delta(a),\ \max_{p\in[\beta,1]}(1-p)(1-e^{-mp})\right\}.} \tag{35}$$

Theorem 4 (Policy attainment and optimized-flat dominance)**.** *For every no-entry parameter tuple with $m > 0$ and $\alpha > 0$, the announced policy value in (11) is attained. If $\beta \geq 1/2$, every optimal menu is a strict menu*

$$(p_1^*, p_2^*) = (P_\delta(a^*), Q(a^*)), \qquad a^* \in \arg\max_{a\in[0,\beta]} J_\delta(a) \subset (0,\beta), \tag{36}$$

and

$$D^*_{\alpha,\delta,0}(m) > F_0^*(m). \tag{37}$$

No uniqueness of a^* is asserted.

Equation (35) is a maximum of continuous functions on compact intervals, so it proves attainment directly. In the no-entry flat problem,

$$F_m(p) = (1-p)(1-e^{-mp})$$

is strictly concave and its unique optimizer is

$$p_F(m) = \frac{1+m-W_0(e^{1+m})}{m} < \frac{1}{2}. \tag{38}$$

For $\beta \geq 1/2$, $p_F < \beta$, and (22) applied at p_F proves (37), including the equality case $\beta = 1/2$. The boundary portions of (35) do not beat F_0^*, whereas (37) does, so every global optimizer is interior and strict.

The driver discount also yields a clean comparative static. Write $R_1(a) = S(a)/[\beta h(a)]$, so that $(1-P_\delta)(P_\delta - a) = \delta R_1(a)$. For $a \in (0, \beta)$,

$$\frac{\partial P_\delta(a)}{\partial \delta} = \frac{R_1(a)}{1+a-2P_\delta(a)} > 0, \qquad \frac{\partial J_\delta(a)}{\partial \delta} = -(1-e^{-ma})\frac{\partial P_\delta(a)}{\partial \delta} < 0. \tag{38a}$$

Thus $D^*_{\alpha,\delta,0}(m)$ is weakly decreasing in δ: more patient incumbents require a higher first payment to implement the same cutoff and continuation coverage. This is a completion comparative static, not a welfare ranking.

7.4 The full patience region

Fix (m, α, δ) and make patience explicit in $D(m; \beta)$. For a fixed strict menu $p < q$, its value equals $F_m(p)$ for $\beta \leq q$ and is strictly increasing for $\beta > q$. Indeed, increasing β raises the active-rescue coefficient in (17), shifts the unique cutoff strictly left (or to zero), and then raises completion by (20)–(21).

The scalar representation is jointly continuous in patience. To use a fixed choice set, write $a = \beta x$. The solution of

$$e^{ky} - 1 = k\{\beta(1-x) - y\}$$

is continuous in (β, x). At $x = 0$, $h(0) = m$ and $P_{\beta,\delta}(0) = p_z(\beta)$ has a strictly positive discriminant; at $x = 1$, $P_{\beta,\delta}(\beta) = Q_\beta(\beta) = \beta$ and $J_{\beta,\delta}(\beta) = (1-\beta)(1-e^{-m\beta})$. These removable extensions and the elementary maximum theorem make $D(m; \beta)$ continuous.

Define

$$\beta_c(m, \alpha, \delta) = \inf\{\beta \in (0,1) : D^*_{\alpha,\delta,0}(m; \beta) > F_0^*(m)\}. \tag{39}$$

Theorem 5 (Patience threshold)**.** *For $m > 0$, $0 < \alpha \leq 1$, and $0 < \delta \leq 1$, the strict-gain region is the single upper interval*

$$D^*_{\alpha,\delta,0}(m; \beta) > F_0^*(m) \quad \Longleftrightarrow \quad \beta > \beta_c(m, \alpha, \delta). \tag{40}$$

Moreover

$$0 < -\frac{1}{m}\log(1 - F_0^*(m)) \leq \beta_c(m, \alpha, \delta) \leq p_F(m) < \frac{1}{2}. \tag{41}$$

For fixed (m, α, δ), $D(m; \beta) = F_0^*(m)$ through the threshold and is strictly increasing above it. The exact scalar test is

$$\beta > \beta_c(m, \alpha, \delta) \quad \Longleftrightarrow \quad \max_{a \in [0, \beta]} J_{\beta,\delta}(a) > F_0^*(m). \tag{42}$$

When $\alpha = 0$, the gain region is empty.

Pointwise monotonicity in (38a) also nests the thresholds:

$$0 < \delta_1 < \delta_2 \le 1 \quad \Longrightarrow \quad \beta_c(m, \alpha, \delta_1) \le \beta_c(m, \alpha, \delta_2). \tag{42a}$$

Strict inequality is not asserted globally.

The lower bound follows because an active menu has $q < \beta$ and completion requires an initial incumbent with cost at most q, giving $M < 1 - e^{-m\beta}$. The upper bound follows from a strict menu at p_F when $\beta > p_F$. If gain holds at one patience level, attainment supplies an optimal strict menu; reusing it at any larger β and applying the fixed menu monotonicity gives strict gain and strict value increase. Continuity makes the gain interval open.

8 Market thickness: topology and endpoint rates

Fix $0 < \alpha \le 1$, $0 < \beta < 1$, $0 < \delta \le 1$, and $\gamma = 0$. Define the optimized value of commitment by

$$V_{\alpha,\delta,0}(m) = D^*_{\alpha,\delta,0}(m) - F^*_0(m). \tag{43}$$

The outer representation (35), rather than a local-menu derivative, is the appropriate object because the optimal menu moves with thickness.

For the thin limit, define on $a \in [0, \beta]$

$$P_0^\delta(a) = \frac{1 + a - \sqrt{(1-a)^2 - \delta\alpha(\beta - a)^2/\beta}}{2}, \qquad H_0^\delta(a) = a[1 - P_0^\delta(a)] + \frac{\alpha(\beta - a)^2}{4\beta}, \tag{44}$$

and $G_{\alpha,\beta,\delta} = \max_{a\in[0,\beta]} H_0^\delta(a)$.

Theorem 6 (Thickness topology and exact endpoint regimes)**.** *For fixed $(\alpha, \beta, \delta) \in (0, 1] \times (0, 1) \times (0, 1]$:*

(i) $D^*_{\alpha,\delta,0}(m)$, $F^*_0(m)$, and $V_{\alpha,\delta,0}(m)$ are continuous on $(0, \infty)$. The scalar cutoff and flat-price argmax correspondences are nonempty, compact valued, and upper hemicontinuous.

(ii) If $\beta > 1/2$ and $\delta < 1$, then

$$V_{\alpha,\delta,0}(m) \sim \left(G_{\alpha,\beta,\delta} - \frac{1}{4}\right) m, \qquad G_{\alpha,\beta,\delta} > \frac{1}{4}. \tag{45}$$

At the no-discount boundary $\delta = 1$, let $a_0 \in (0, 1/2)$ be the unique solution of

$$\alpha(\beta - a_0)^2 = \beta(1 - 2a_0).$$

Then

$$V_{\alpha,1,0}(m) \sim \frac{1 - 2a_0}{16} m^2. \tag{46}$$

(iii) If $\beta = 1/2$ and $\delta < 1$, then

$$V_{\alpha,\delta,0}(m) \sim \frac{\alpha(1-\delta)}{512\{1 - \alpha(1-\delta)/2\}} m^3. \tag{47}$$

At $\delta = 1$, the leading cancellation persists one order longer:

$$V_{\alpha,1,0}(m) \sim \frac{\alpha}{2048} m^4. \tag{48}$$

(iv) If $\beta < 1/2$, there exists $m_0(\alpha, \beta, \delta) > 0$ such that

$$D^*_{\alpha,\delta,0}(m) = F^*_0(m), \qquad V_{\alpha,\delta,0}(m) = 0, \qquad 0 < m < m_0(\alpha, \beta, \delta). \tag{49}$$

(v) As $m \to \infty$,

$$1 - D^*_{\alpha,\delta,0}(m) \sim \frac{\log\log m}{m}, \quad 1 - F^*_0(m) \sim \frac{\log m}{m}, \quad V_{\alpha,\delta,0}(m) \sim \frac{\log m}{m}. \tag{50}$$

For each fixed positive (α, β), the estimates establishing (50) are uniform over the full policy/cutoff domain and uniformly over $\delta \in (0, 1]$. They are not uniform over primitive sequences with $\alpha \downarrow 0$ or $\beta \downarrow 0$.

Equations (45)–(48) show a singular boundary: the limits $m \downarrow 0$ and $\delta \uparrow 1$ do not commute at the first nonzero order. Even slight incumbent impatience creates a first-order screening gain when $\beta > 1/2$ and a cubic gain at $\beta = 1/2$; exact patience removes those terms.

Here is the proof architecture. Let $k = \alpha m$ and let $z_m(a)$ solve

$$z_m(a) + \log[1 + z_m(a)] = k(\beta - a), \qquad y_m(a) = z_m(a)/k.$$

Then

$$Q_m(a) = \beta - y_m(a), \quad T_m(a) = \frac{k y_m(a)^2}{\beta[1 + k y_m(a)]}, \quad B^\delta_m(a) = \delta \frac{a}{e^{ma} - 1} T_m(a), \tag{51}$$

with $a/(e^{ma} - 1) = 1/m$ at zero, and

$$P^\delta_m(a) = \frac{1 + a - \sqrt{(1 - a)^2 - 4B^\delta_m(a)}}{2}. \tag{52}$$

The scalar objective admits the useful exact form

$$J^\delta_m(a) = [1 - P^\delta_m(a)](1 - e^{-ma}) + T_m(a)e^{-ma}. \tag{53}$$

The strict envelope bound keeps the discriminant positive. Joint continuity and Berge's maximum theorem prove part (i).

Uniformly on $[0, \beta]$, $P^\delta_m \to P^\delta_0$ and $J^\delta_m/m \to H^\delta_0$. If $\beta > 1/2$ and $\delta < 1$, the unique $a \in (0, 1/2)$ satisfying $P^\delta_0(a) = 1/2$ gives

$$H^\delta_0(a) = \frac{1}{2}\left[a + \frac{1/2 - a}{\delta}\right] > \frac{1}{4},$$

which proves the strict coefficient in (45); uniform convergence gives its exact value as the maximum in (44). For $\delta = 1$, the first-order term ties the flat benchmark, and the uniform second-order argmax expansion gives (46).

At $\beta = 1/2$, every thin optimizer localizes as $a = 1/2 - cm$. Put $A_\delta = 1 - \alpha(1 - \delta)/2$. Uniformly for bounded c,

$$J^\delta_m(1/2 - cm) = \frac{m}{4} - \frac{m^2}{16} + m^3\left\{\frac{1}{96} + \frac{c}{8} - A_\delta c^2\right\} + O(m^4). \tag{54}$$

The maximizing rescaled cutoff is $c^* = 1/(16A_\delta)$, while the flat m^3 coefficient is $11/768$. Their difference is exactly the coefficient in (47). When $\delta = 1$ it vanishes; the fourth-order expansion yields (48). In particular, uniformly for bounded c,

$$\begin{aligned} J_m^1(1/2-cm) &= \frac{m}{4}-\frac{m^2}{16}+m^3 f_2(c)+m^4 f_3(c)+o(m^4),\\ f_2(c) &= \frac{11}{768}-\left(c-\frac{1}{16}\right)^2, \\ f_3(c) &= -\frac{1}{768}-\frac{c}{24}+\frac{c^2}{4}+2\alpha c^3, \end{aligned} \tag{54a}$$

whereas

$$F_0^*(m)=\frac{m}{4}-\frac{m^2}{16}+\frac{11}{768}m^3-\frac{3}{1024}m^4+o(m^4). \tag{54b}$$

The strict quadratic loss in f_2 and local Lipschitz control of f_3 evaluate the perturbed maximum at $c=1/16$ to fourth order, and $f_3(1/16)=-3/1024+\alpha/2048$. Subtracting (54b) proves (48). If $\beta<1/2$, the bound

$$H_0^\delta(a)\le a(1-a)+\frac{(\beta-a)^2}{4\beta}\le\beta(1-\beta)<\frac{1}{4}$$

is uniform on $[0,\beta]$. Since the flat optimizer converges to 1/2, the flat policy wins exactly on a nonempty initial interval, proving (49).

For the thick limit put $x=ma$, $p=P_m^\delta(a)$, and $T=T_m(a)$. Equation (53) gives the same exact nonnegative loss decomposition for every fixed $\delta>0$:

$$\boxed{1-J_m^\delta(a)=p(1-e^{-x})+(1-T)e^{-x}.} \tag{55}$$

The feasible choice $x=\log\log m$ and the policy-uniform lower bound obtained by splitting at $Q_m(a)=\beta/2$ give the first rate in (50). The exact flat first-order condition gives the second, and subtraction gives the third.

Corollary 1 (A best finite thickness, without a shape theorem)**.** *For every fixed positive* (α,β,δ) *in the maintained ranges, there exists* $m^*\in(0,\infty)$ *such that*

$$V_{\alpha,\delta,0}(m^*)=\max_{m>0}V_{\alpha,\delta,0}(m)>0. \tag{56}$$

Strict single-peakedness is false when $\beta<1/2$ *because of (49). For* $\beta\ge 1/2$*, strict single-peakedness remains open for every fixed* $\delta\in(0,1]$*.*

Continuity, endpoint vanishing, and eventual positivity in (50) localize a global maximizer to a compact subinterval. No uniqueness of m^* is claimed.

9 A public-supply benchmark

The Poisson model makes realized local supply latent. To separate that information friction from assignment competition, fix instead a deterministic incumbent count $n\ge1$ observed by every agent. We evaluate one announced menu conditional on that fixed n; the exercise removes rival-count uncertainty without introducing state-contingent pricing. Keep the timing, uniform private costs, survival, and no-entry assumption. For a conjectured cutoff a, a focal acceptor faces $X\sim\mathrm{Bin}(n-1,a)$ accepting rivals, so her assignment share is

$$s_n(a):=\mathbb{E}\left[\frac{1}{1+X}\right]=\frac{1-(1-a)^n}{na},\qquad s_n(0)=1. \tag{57}$$

Universal rejection occurs against all of her rivals with probability $(1-a)^{n-1}$. Conditional on that event, each rival has cost uniform on $(a,1)$. If terminal payment p_j is selected, a rival is both surviving and eligible with probability

$$\theta_j(a) = \frac{\alpha(p_j - a)}{1-a};$$

hence a terminal acceptor's assignment share is $s_n(\theta_j)$ and terminal coverage is

$$C_{j,n}(a) = 1 - [1-\theta_j(a)]^n = n\theta_j(a)s_n(\theta_j(a)). \tag{58}$$

Let $\eta_j(a)$ be the posting-conditional mass of riders who select terminal payment p_j after failure. At a positive interior cutoff, indifference of the cutoff driver is

$$s_n(a)(p_1 - a) = \delta(1-a)^{n-1}\alpha\sum_j \eta_j(a)s_n(\theta_j(a))(p_j - a). \tag{59}$$

Substitution of (58)–(59) into unconditional completion gives the exact finite-population counterpart of (20):

$$\boxed{M_n(p_1,p_2;a) = (1-p_1)ns_n(a)\left[a + \frac{p_1 - a}{\delta}\right].} \tag{60}$$

Proposition 3 (Public n and local announced rescue)**.** *Fix $n \geq 1$, $0 < \alpha \leq 1$, $0 < \delta \leq 1$, and $p \in (0,\beta)$. For every sufficiently small $\varepsilon > 0$, the menu $(p, p+\varepsilon)$ has a unique symmetric cutoff, and it satisfies*

$$a_\varepsilon = p - \kappa_n\varepsilon + o(\varepsilon), \qquad \kappa_n = \frac{\delta(1-p)^{n-1}\alpha\rho(p)}{s_n(p) - \delta(1-p)^{n-1}\alpha\rho(p)} > 0. \tag{61}$$

Its right derivative of unconditional completion is

$$\left.\frac{dM_n}{d\varepsilon}\right|_{0+} = (1-p)n\kappa_n\left\{\frac{1-\delta}{\delta}s_n(p) - p\,s_n'(p)\right\}. \tag{62}$$

This derivative is strictly positive for every $n \geq 1$ when $\delta < 1$. At $\delta = 1$, it is strictly positive for $n \geq 2$ and exactly zero for $n = 1$.

The denominator in (61) is positive because $s_n(p) \geq (1-p)^{n-1}$ and $\delta\alpha\rho(p) < 1$. Moreover

$$s_n(a) = \frac{1}{n}\sum_{r=0}^{n-1}(1-a)^r,$$

which is strictly decreasing for $n \geq 2$ and constant for $n = 1$. Proposition 3 rules out the claim that an unobserved realized count is necessary for the rescue gain. The first term in (62) is pure intertemporal compensation and survives at $n = 1$; the second is assignment competition and is positive exactly when $n \geq 2$. Latent supply changes posterior inference and the global policy problem, but neither local wedge. If $\alpha = 0$, then $\kappa_n = 0$ and the derivative in (62) is zero for every n.

10 Managerial interpretation and institutional scope

The rescue payment is valuable because it is a commitment, not because a platform can mechanically raise pay after observing failure. Announcing the second payment before drivers act makes delay incentive compatible with the policy and therefore forces the platform to price the delay externality. In the no-entry model, the exact fixed-p_1 solution says more than "raise the payment a little." Low first payments optimally generate reject-all and use the unique rescue-envelope maximizer; intermediate first payments use the unique tangent rescue that implements the smallest feasible cutoff; once the first payment reaches rider patience β, rescue is irrelevant.

Three operational implications follow. First, the relevant demand-side state is rider patience, not only nominal willingness to pay. The exact threshold $\beta_c(m, \alpha, \delta)$ identifies when commitment beats the best flat payment, while the simple condition $\beta \geq 1/2$ guarantees gain at every finite thickness. Second, market thickness changes the form of the optimal policy. In a very thick market the dynamic policy closes the request with loss of order $\log\log m/m$, compared with $\log m/m$ under the best flat payment. Yet the absolute difference still vanishes, so a large percentage reduction in the small residual failure probability should not be confused with a large unconditional completion gain. Third, publishing realized local supply is not enough to eliminate the local mechanism. Driver impatience already makes rescue productive with one publicly known incumbent; assignment competition adds a second gain when there are two or more.

The maintained transfer is paid by the rider to the selected driver, and the objective is completion. This maps most directly to bidding, tipping, top-up, freight, and task platforms. A platform-funded bonus, profit maximization, or a bonus budget introduces a real resource cost and is not covered by the completion theorem. Likewise, simultaneous acceptance with uniform selection is a reduced form for a broadcast response window; a first-response race would add endogenous response effort and is not claimed to be equivalent.

The equilibrium scope also matters. The arbitrary-menu analysis covers anonymous symmetric pure-cutoff WPBE and takes a conservative minimum only inside that class. We do not claim robustness to asymmetric or mixed WPBE. Fresh entry is fully incorporated in the equilibrium characterization and flat benchmark, but the global menu solution is for $\gamma = 0$. Finally, the thickness theorem proves existence, endpoint rates, and one exact failure of strict single-peakedness; it does not prove a unique best thickness when $\beta \geq 1/2$.

11 Conclusion

An announced rescue payment changes both continuation coverage and current acceptance. Solving those effects jointly reveals a tractable global design problem. In the no-entry baseline, every menu has a unique cutoff, the conditional rescue problem is a concave envelope, and the full two-price problem reduces to one continuous scalar maximum. Commitment strictly beats the optimized flat payment above an endogenous patience threshold and always does so at $\beta \geq 1/2$. Optimized thickness effects have distinct thin-market regimes and matching thick-market rates, while strict single-peakedness is either false or still unproved. The public-n benchmark decomposes the local gain into intertemporal compensation and assignment competition, assigning latent supply the narrower role of shaping inference and global design.

A Proofs for the equilibrium characterization

Proof of Lemma 1. Poisson marking splits the incumbent process into costs below and above a, with independent counts of means ma and $m(1-a)$. Thus first-period coverage is $1-e^{-ma}$. Under Palm conditioning a focal driver's accepting rivals form a Poisson random variable Z of mean ma, and

$$\mathbb{E}\big[(1+Z)^{-1}\big] = \int_0^1 \mathbb{E}[t^Z]dt = \int_0^1 e^{-ma(1-t)}dt = \phi(ma).$$

If the focal driver waits, continuation is reached only if every rival has cost above a, an event of probability e^{-ma}. Conditional on no point below a, the above-a Poisson process is unchanged. Independent survival thins its eligible interval $(a, p_j]$ to intensity $m\alpha(p_j - a)$, while fresh eligible drivers contribute $m\gamma p_j$. Superposition gives λ_j in (1), terminal coverage C_j, and the same integral identity gives terminal assignment share $\phi(\lambda_j)$. □

Proof of Proposition 1. Given a posted request, failure is independent of v, so the rider retains the posterior $U[p_1, 1]$. Comparing the two affine continuation payoffs gives v^M and the masses in (2); the stated priority rule assigns the zero-payoff and positive repeat–rescue ties. Before posting, a type $v < p_1$ has strictly negative payoff from immediate completion and cannot obtain positive delayed surplus because $\beta v - p_j < v - p_1 < 0$. A type $v \geq p_1$ obtains a nonnegative immediate payoff whenever a driver accepts and otherwise can abandon. The posting rule therefore is exactly $v \geq p_1$.

For $c \leq p_1$, equations (4)–(5) give the affine margin (6). Since $\eta^R + \eta = \rho(p_1) < 1$ whenever it is positive,

$$B(a) \geq \phi(ma) - \delta e^{-ma} \geq \phi(ma) - e^{-ma} > 0 \quad (a > 0),$$

and at $a = 0$, $B(0) \geq 1 - \delta\alpha\rho(p_1) > 0$. Thus $A - W$ is strictly decreasing in cost; types above p_1 never prefer the negative current surplus to waiting. A symmetric best response is consequently a genuine cutoff, and evaluation at its cutoff type gives (7).

At $a = p_1$, $f_m(p_1) \leq 0$. If $f_m(0) \leq 0$, the lower boundary is an equilibrium. If $f_m(0) > 0$ and $f_m(p_1) < 0$, continuity and the intermediate value theorem give an interior zero; if $f_m(p_1) = 0$, the upper boundary is an equilibrium. The aggregate continuation payoff is continuous across a rider switch because tied actions have equal coverage-weighted utility; when all continuation coverage is zero the driver's wait payoff is zero. Hence f_m is continuous. The same argument shows that any convergent sequence of eligible boundary points or zeros has an eligible limit, so the nonempty equilibrium set is closed in $[0, p_1]$ and therefore compact. The separately stated $p_1 = 0$ and $(1, 1)$ conventions close the remaining endpoints. Formula (9) is continuous on the compact equilibrium set, making the minimum in (10) well defined. □

Proof of Proposition 2. For $a < p$, the cutoff margin is $(p - a)$ times the bracket displayed below (13). If $a > 0$, then $\phi(ma) > e^{-ma}$ and $\delta\alpha\rho(p)\phi(m\{\alpha(p-a) + \gamma p\}) \leq 1$, so the bracket is strictly positive. At $a = 0$, strictness follows from $\rho(p) < 1$. Hence no cutoff below p is sustainable, while $a = p$ makes the cutoff type indifferent and is the unique numeric cutoff. At that cutoff no incumbent with cost below p remains. Terminal completion can only come from fresh drivers of intensity γmp, and the mass of posting riders with nonnegative delayed surplus is $[1 - p/\beta]^+/(1-p)$. Adding initial and terminal completion yields (12). The continuous objective on $[0, 1]$ attains the flat maximum. □

Proof of Theorem 1. Put $d = p - a$. Under $(p, p + \varepsilon)$ the two terminal intensities are

$$\lambda_1 = m(\alpha d + \gamma p), \qquad \lambda_2 = \lambda_1 + m(\alpha + \gamma)\varepsilon. \tag{A1}$$

If $\eta_\varepsilon(a)$ is the rescue mass, interior cutoff indifference is

$$\begin{aligned}\phi(ma)d = \delta\alpha e^{-ma}\{[\rho - \eta_\varepsilon]\phi(\lambda_1)d \\ + \eta_\varepsilon\phi(\lambda_2)(d+\varepsilon)\}.\end{aligned} \tag{A2}$$

Rearrange and use $\phi(\lambda_j) \leq 1$, $\eta_\varepsilon \leq \rho$, and $\phi(ma) \geq e^{-ma}$. The coefficient on d is at least $e^{-ma}(1 - \delta\alpha\rho) > 0$, which gives (L4). The same weak boundary inequality excludes a reject-all cutoff for small ε; the upper cutoff already obeys the bound. Thus every equilibrium is localized, not only a selected branch.

On the compact rescaling $y = d/\varepsilon$, (A1) gives

$$\frac{C_2 - C_1}{\varepsilon} \to mE(\alpha + \gamma), \qquad C_2 \to 1 - E.$$

The rider switch therefore converges uniformly to $\bar{v}$ in (L2), and the rescue mass converges to $\eta^0 > 0$. Divide (A2) by ε and pass to the limit:

$$[\sigma - \delta R\alpha\rho\ell]y - \delta R\alpha\eta^0\ell = 0. \tag{A3}$$

Its slope is at least $\sigma - R > 0$. The rescaled margin converges in C^1 to this affine function, so it has a unique zero for all small ε; (A3) yields (L3).

It remains to establish the strict sign in (L6). First set $\delta = 1$ and call the resulting expression T_1. The case $\gamma = 0$ gives $T_1 = \alpha(\sigma - R) > 0$. For $\gamma > 0$, set

$$u = e^x - 1, \quad t = 1 - E, \quad A = \alpha + \gamma, \quad \Delta = 1 - E(1+\gamma), \quad B = 1 - \rho\Delta.$$

Since $\sigma = Ru/x$ and $\ell = t/(\gamma x)$,

$$T_1 = \frac{R}{\gamma x}Q, \qquad Q = E\gamma Au - \alpha tB. \tag{A4}$$

If $\Delta \leq 0$, then $B \leq E(1+\gamma)$, $\gamma u > t$, and $A \geq \alpha(1+\gamma)$, so $Q > 0$. If $\Delta > 0$ and $\gamma \leq 1$, convexity gives $t \leq E\gamma u$ and $B \leq 1$, whence $Q \geq \alpha(E\gamma u - t) + E\gamma^2 u > 0$.

Finally suppose $\Delta > 0$ and $\gamma > 1$. Activity implies the weaker strict bound

$$\rho > \frac{t}{t + xEA}.$$

It implies

$$Q \geq \frac{E}{t + xEA}I(\alpha), \quad I(\alpha) = \gamma Au(t + xEA) - \alpha t\{xA + t(1+\gamma)\}. \tag{A5}$$

The coefficient on α^2 in I is $x(\gamma Eu - t) < 0$, because $e^{\gamma x} - 1 > \gamma(e^x - 1)$. Hence the minimum of this concave quadratic on $[0, 1]$ is at an endpoint. Clearly $I(0) > 0$. At $\alpha = 1$, write $H = t + xE(1+\gamma)$ and $d = \gamma u - t$. Then

$$\frac{I(1)}{1+\gamma} = dH - tx\Delta. \tag{A6}$$

Let $w = \gamma x$. Since $\Delta > 0$, $w > \log(1+\gamma)$; moreover

$$d > \frac{\gamma x^2}{2} = \frac{xw}{2}, \qquad \Delta = 1 - (1+\gamma)e^{-w} < \frac{w}{2}.$$

The last inequality follows by minimizing $w/2 - 1 + (1+\gamma)e^{-w}$ on $[\log(1+\gamma), \infty)$. Thus $d > x\Delta$ and, since $H > t$, (A6) is positive. This proves $Q > 0$ in all cases and hence $T_1 > 0$. For general $0 < \delta \le 1$,

$$T_\delta = T_1 + (1-\delta)R\alpha\ell\{1 - \rho + \rho E(1+\gamma)\} > 0.$$

Along the cutoff branch, write continuation coverage as $S_\varepsilon = \rho C_1 + \eta_\varepsilon(C_2 - C_1)$. Direct differentiation at zero gives

$$S_0' = mE\{\rho\alpha\kappa + \eta^0(\alpha+\gamma)\}.$$

Differentiating (9), substituting (L3), and collecting terms yields exactly (L5)–(L6). All remaining factors are positive, completing the proof. □

B Proofs for global no-entry design

Proof of Theorem 2. The cancellation leading to (16) follows by substituting the rider switch type into the sum of repeat and rescue coverage. Inequality (18) excludes the repeat branch and its kink from the zero set. Every zero is therefore a smooth active-rescue zero. With the notation preceding (19), its root equation, $B - b = q - \beta p > 0$, and $\delta \le 1$ imply $u < C(z)/m$. Consequently

$$\frac{1}{u} - \frac{k}{e^{kz} - 1} > m\frac{1 - \alpha e^{-kz}}{1 - e^{-kz}} \ge m > \frac{h'(a)}{h(a)},$$

where $h'(a)/h(a) < m$ follows from $h(a) = \int_0^m e^{ta}dt$. Thus (19) is strictly negative and every zero crosses downward. Also $\mathcal{D}^\delta_{p,q}(p) = -\delta(\beta - q)C(q-p)/B_p < 0$. A continuous function whose every zero is a strict downward crossing has at most one zero. The endpoint signs give exactly the two cases stated in the theorem, including the equality case at zero. Flat and repeat-only menus use the strict flat inequality and have cutoff p; if $p \ge \beta$, no positive mass continues; if $\alpha = 0$, waiting has zero payoff; and $p = 0$ has only the lower numeric cutoff. These observations cover every no-entry menu. □

Proof of Theorem 3. For a target cutoff a, active-rescue implementability is exactly (31). The strict flat inequality makes its right side larger than $A(a,p)$, so an implementer is strictly inside the rescue region. The function $A(a,q)$ is strictly concave because

$$A_{qq}(a,q) = -e^{-k(q-a)}[2k + k^2(\beta - q)] < 0,$$

and its unique maximizer is $Q(a)$. At $a = 0$, reject-all is implementable iff $A(0,q) \ge \beta(1-p)mp/\delta$. Its largest possible left side is S_0, so the condition is feasible exactly when $p \le p_z$; the strict flat inequality at $(a,p) = (0, Q_0)$ gives $0 < p_z < Q_0 < 1/2$.

If $p \le p_z$, any positive-cutoff completion is below $(1-p)mp/\delta \le S_0/\beta$, whereas reject-all completion is $A(0,q)/\beta \le S_0/\beta$. Strict concavity makes $q = Q_0$ the unique maximizer, including at $p = p_z$. If $p_z < p < \beta$, maximize A pointwise in q and define the envelope margin

$$\overline{\mathcal{D}}_p(a) = (p-a)h(a) - \delta S(a)/[\beta(1-p)].$$

It is positive at zero and negative at p. The same logarithmic derivative argument as in Theorem 2 shows that every envelope zero crosses downward, hence (30) has exactly one root $a^*(p)$. Any menu implementing a satisfies $A(a,q) \le S(a)$, so its unique cutoff cannot lie below $a^*(p)$. Completion in (20) decreases with a positive cutoff; therefore the tangent menu $q = Q(a^*)$ is the unique maximizer. For $p \ge \beta$, no positive mass uses rescue and all $q \ge p$ are outcome-equivalent to the flat payment. This proves all three regimes and attainment. □

Proof of Theorem 4. Let $g_a(p) = (1-p)(p-a)$. Equation (26) implies $Q(a) < (1+a)/2$, so g_a is strictly increasing on $(a, Q(a))$. The strict flat inequality gives $g_a(Q(a)) > R_\delta(a)$, while $g_a(a) = 0 < R_\delta(a)$; hence (32) has exactly one root in that interval, the lower branch (33). If two distinct cutoffs had the same value of P_δ, the same fixed-p envelope would have two downward-crossing zeros. This also excludes $P_\delta(a) = P_\delta(0)$ for $a > 0$. Thus the continuous map P_δ is strictly increasing from $[0, \beta]$ onto $[p_z, \beta]$.

The three fixed-price regimes now give exactly (35). Both objectives are continuous on fixed compact intervals, so each maximum and the original policy maximum are attained. The flat objective is strictly concave; its first-order condition is $e^{mp} - 1 = m(1-p)$, whose unique root satisfies $p_F < 1/2$. For any $p < q < \beta$, Theorem 2 gives a cutoff below p; equations (20)–(21) then make completion strictly larger than $F_m(p)$. If $\beta \geq 1/2$, apply this comparison at $p_F < \beta$. The reject-all boundary and $p \geq \beta$ flat region do not beat F_0^*, whereas the strict comparison does, so every global maximizer is a strict interior menu parameterized as in (36). No step requires uniqueness of the outer argmax. □

Proof of Theorem 5. First fix a strict menu $p < q$. If $\beta \leq q$, rescue has no positive-mass use and the strict flat inequality forces cutoff p, so the menu value is $F_m(p)$. If $q < \beta_1 < \beta_2$, the coefficients $(\beta - z)/[\beta(1-p)]$, $z \in \{p, q\}$, weakly increase in β and the active rescue coefficient increases strictly. At the β_1 cutoff this strictly lowers the scalar margin for β_2. Global downward crossing therefore moves the β_2 cutoff strictly left, unless it is already zero. Equations (20)–(21) then strictly increase completion. At a zero cutoff the value is $(1 - q/\beta)C(q)$ and has positive derivative. As $\beta \downarrow q$, any cutoff limit below p would contradict (18), so the menu value is continuous at q.

Continuity of the optimized value follows from the fixed coordinate $a = \beta x$, $x \in [0, 1]$. The unique solution of $e^{ky} - 1 = k\{\beta(1-x) - y\}$ and all objects $Q, S, P_{\beta,\delta}, J_{\beta,\delta}$ are jointly continuous, with the removable extensions described before (39). The restricted flat term is jointly continuous after writing $p = \beta + (1-\beta)s$. The maximum theorem applied to these fixed compact sets makes $D(m; \beta)$ continuous.

Flat policies give $D \geq F_0^*$. If strict gain holds at β_1, attainment supplies an optimal strict menu with $q < \beta_1$. Reusing it at any $\beta_2 > \beta_1$ and the preceding fixed-menu result gives strict gain and a strictly larger optimized value. Thus the gain set is an open upper interval and has the form (40). An active menu has $q < \beta$ and completion at most $1 - e^{-mq} < 1 - e^{-m\beta}$; this proves the positive lower bound in (41). If $\beta > p_F$, a strict menu at first payment p_F beats the flat optimum by (22), proving the upper bound. Finally, the second maximum in (35) never exceeds F_0^*, so strict gain is equivalent to the scalar test (42). If $\alpha = 0$, no incumbent can complete after waiting and the dynamic and flat values coincide. □

C Proofs for thickness and the public-supply benchmark

Proof of Theorem 6. Equations (51)–(53) are the fixed-coordinate form of (26)–(34). The inverse of $z \mapsto z + \log(1+z)$ is continuous. Since multiplying the old envelope term by $0 < \delta \leq 1$ only enlarges the lower-root discriminant, the strict envelope bound keeps that discriminant positive. Thus J_m^δ is jointly continuous on every compact m-interval times $[0, \beta]$. Berge's maximum theorem applied to (35) proves part (i).

For $m \downarrow 0$, series inversion, uniformly in $a \in [0, \beta]$, gives

$$y_m(a) = \frac{\beta - a}{2} + O(m), \qquad \frac{S_m(a)}{m} = \frac{\alpha(\beta - a)^2}{4} + O(m).$$

Consequently $P_m^\delta \to P_0^\delta$ and $J_m^\delta/m \to H_0^\delta$ uniformly. If $\beta > 1/2$ and $\delta < 1$, continuity and the endpoint signs give an $a \in (0, 1/2)$ with $P_0^\delta(a) = 1/2$. The value displayed after (53) is then strictly above $1/4$. Hence $G_{\alpha,\beta,\delta} > 1/4$, and uniform convergence together with $F_0^*(m)/m \to 1/4$ proves (45).

If $\delta = 1$, the limiting objective is $P_0^1(a)[1 - P_0^1(a)]$. Its unique maximum for $\beta > 1/2$ occurs at $P_0^1(a_0) = 1/2$, equivalently $\alpha(\beta - a_0)^2 = \beta(1 - 2a_0)$. Uniform second-order expansion gives

$$D^*_{\alpha,1,0}(m) = \frac{m}{4} - \frac{a_0}{8}m^2 + o(m^2), \qquad F_0^*(m) = \frac{m}{4} - \frac{1}{16}m^2 + o(m^2),$$

which proves (46).

Now let $\beta = 1/2$. The limiting objective reaches $1/4$ only at $a = 1/2$, with a quadratic loss away from that endpoint. A uniform comparison with the endpoint therefore localizes every optimizer as $a = 1/2 - cm$ with bounded c. Expanding (51)–(53) on this compact rescaling gives (54), uniformly in c. Its strictly concave quadratic is maximized at $c^* = 1/(16A_\delta)$. Since

$$F_0^*(m) = \frac{m}{4} - \frac{m^2}{16} + \frac{11}{768}m^3 + O(m^4),$$

the difference of the cubic coefficients is

$$\frac{1}{256A_\delta} - \frac{1}{256} = \frac{\alpha(1-\delta)}{512A_\delta},$$

proving (47). At $\delta = 1$ this difference is zero. The next uniform order is given by (54a)–(54b). The strict quadratic loss in f_2 and local Lipschitz control of f_3 imply

$$\max_c\{f_2(c) + mf_3(c) + o(m)\} = f_2(1/16) + mf_3(1/16) + o(m).$$

Since $f_3(1/16) = -3/1024 + \alpha/2048$, subtracting the flat expansion proves (48) without assuming a higher-order expansion of the optimizer itself.

If $\beta < 1/2$, $P_0^\delta(a) \ge a$ and $\alpha \le 1$ imply the bound stated after (54). Its right side is increasing on $[0, \beta]$ and ends at $\beta(1 - \beta) < 1/4$. Uniform convergence and $p_F(m) \to 1/2 > \beta$ make the flat term the exact winner for all sufficiently small m, proving (49).

For $m \to \infty$, (55) is exact. With $x = \log\log m$ and $a = x/m$, the tangent construction gives $q = O(\log m/m)$, $p - a = O(\delta a e^{-x})$, and $1 - T = O(\log m/m)$, yielding the feasible upper loss $(1 + o(1))\log\log m/m$. For the policy-uniform lower bound, if $Q_m(a) \le \beta/2$, then

$$m[1 - J_m^\delta(a)] \ge x(1 - e^{-x}) + \frac{\log(1 + \alpha m\beta/2)}{\alpha\beta}e^{-x}.$$

Its minimum is asymptotic to $\log\log m$. If $Q_m(a) > \beta/2$, the envelope bound makes the first loss in (55) larger. This proves the dynamic rate. The flat first-order condition gives $1 - F_0^*(m) \sim \log m/m$; subtraction proves the value rate. The lower bound uses only $P_m^\delta(a) \ge a$. For the feasible upper construction, $R_\delta = \delta R_1$ implies $P_m^\delta(a) \le P_m^1(a)$, while $T_m(a)$ is independent of δ. Hence the estimate is uniform over $\delta \in (0, 1]$ for each fixed positive (α, β); only sequences with $\alpha \downarrow 0$ or $\beta \downarrow 0$ fall outside this uniformity. □

Proof of Corollary 1. The thin regimes imply $V(m) \to 0$ as $m \downarrow 0$, and (50) gives both $V(m) \to 0$ and $V(m) > 0$ for all sufficiently large finite m. Choose one such point. Endpoint vanishing makes the value outside some compact subinterval smaller than half its value at that point; continuity supplies a positive global maximizer on the compact interval. Equation (49) directly contradicts strict increase near zero when $\beta < 1/2$. No analogous derivative or optimizer-switch control has been proved for $\beta \ge 1/2$. □

Proof of Proposition 3. The binomial identity in (57) follows by integrating $\mathbb{E}[t^X] = (1-a+at)^{n-1}$. Given universal rejection, each rival cost is uniform on $(a,1)$, giving θ_j, (58), and cutoff indifference (59). Conditional on posting, completion is

$$1-(1-a)^n + (1-a)^n \sum_j \eta_j C_{j,n}.$$

Substitute (58) into the second term and use (59); the expression becomes $ns_n(a)[a+(p_1-a)/\delta]$. Multiplication by posting mass $1-p_1$ proves (60).

For an arbitrary nearby interior cutoff put $d = p-a$. Rearranging its indifference condition gives

$$\begin{aligned}[s_n(a) - \delta(1-a)^{n-1}\alpha\{(\rho-\eta)s_n(\theta_1) + \eta s_n(\theta_2)\}]d \\ = \delta(1-a)^{n-1}\alpha\eta s_n(\theta_2)\varepsilon.\end{aligned}$$

Because $s_n(a) \geq (1-a)^{n-1}$, $s_n(\theta_j) \leq 1$, and $\eta \leq \rho$, every equilibrium satisfies

$$0 \leq d \leq \frac{\delta\alpha\rho}{1-\delta\alpha\rho}\varepsilon.$$

The analogous boundary inequality excludes $a = 0$ for small ε; at $a = p$ the cutoff type strictly prefers waiting. Thus all equilibria are interior and uniformly localize at p.

Write $a = p - \kappa\varepsilon + o(\varepsilon)$. Terminal coverage is first order in ε, the rider switch converges uniformly to p/β, and the rescue mass converges to $\rho(p)$. Divide cutoff indifference by ε and take the limit:

$$s_n(p)\kappa = \delta(1-p)^{n-1}\alpha\rho(p)(1+\kappa).$$

The denominator is positive because $s_n(p) \geq (1-p)^{n-1}$ and $\delta\alpha\rho(p) < 1$, yielding (61). Differentiating (60) along the branch yields (62). The rescaled cutoff margin converges in C^1 on the compact localization interval to the displayed affine equation, whose slope is the positive denominator in (61). Existence from the general cutoff argument and this one-crossing limit give local uniqueness. Finally, $s_n(a) = n^{-1}\sum_{r=0}^{n-1}(1-a)^r$ is strictly decreasing iff $n \geq 2$, which proves the sign statement. □

References

D. Acemoglu, M. Mostagir, and A. Ozdaglar. Managing innovation in a crowd. *NBER Working Paper* 19852, 2014. doi:10.3386/w19852.

Y. Aviv and A. Pazgal. Optimal pricing of seasonal products in the presence of forward-looking consumers. *Manufacturing & Service Operations Management*, 10(3):339–359, 2008. doi:10.1287/msom.1070.0183.

J. Bai, H. S. Heese, and M. Tripathy. Hiding in plain sight: Surge pricing and strategic providers. *Production and Operations Management*, 32(12):3837–3855, 2023. doi:10.1111/poms.14064.

C.-H. Chen. Name your own price at Priceline.com: Strategic bidding and lockout periods. *Review of Economic Studies*, 79(4):1341–1369, 2012. doi:10.1093/restud/rds005.

Y. Chen and M. Hu. Pricing and matching with forward-looking buyers and sellers. *Manufacturing & Service Operations Management*, 22(4):717–734, 2020. doi:10.1287/msom.2018.0769.

J. Correa, R. Montoya, and C. Thraves. Contingent preannounced pricing policies with strategic consumers. *Operations Research*, 64(1):251–272, 2016. doi:10.1287/opre.2015.1452.

N. Garg and H. Nazerzadeh. Driver surge pricing. *Management Science*, 68(5):3219–3235, 2022. doi:10.1287/mnsc.2021.4058.

B. Hu, M. Hu, and H. Zhu. Surge pricing and two-sided temporal responses in ride hailing. *Manufacturing & Service Operations Management*, 24(1):91–109, 2022. doi:10.1287/msom.2020.0960.

Y. Qin, H. Yang, and R. Liu. A two-round broadcasting matching mechanism in ride-sourcing markets. *Transportation Research Part C: Emerging Technologies*, 180:105318, 2025. doi:10.1016/j.trc.2025.105318.

D. Sigg, M. Hardt, and C. Mendler-Dünner. Decline now: A combinatorial model for algorithmic collective action. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*, Article 912, 2025. doi:10.1145/3706598.3713966.

H. Sun, R. Teunter, G. Hua, and T. Wu. Taxi-hailing platforms: Inform or assign drivers? *Transportation Research Part B: Methodological*, 142:197–212, 2020. doi:10.1016/j.trb.2020.10.001.

Z. Wu, L. Wang, F. Huang, L. Zhou, Y. Song, C. Ye, P. Nie, H. Ren, J. Hao, R. He, and Z. Sun. A framework for multi-stage bonus allocation in meal delivery platform. In *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, pages 4195–4203, 2022. arXiv:2202.10695.